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360° RAYS: Representation & Analysis of Your Space

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by

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CERTIFICATE

This is to certify that the work presented in this thesis proposal titled ***360° RAYS: Representation and Analysis of Your Space*** by *Bipasha Garg* has been carried out under my supervision and has not been submitted elsewhere for a degree.

Date

Advisor: Kamalakar Karlapalem

Abstract

The aim of a data visualization is to depict the data in an intuitive way but we come across some limitations when the data to be visualized is humongous in terms of dimensions or data points. We, in this research work, aim to tackle this problem of visualizing high-dimensional data in an efficient and intuitive way.

”360° RAYS” is a visualization built to cater to this problem of visualizing high-dimensional data without the loss of information, unlike in dimensionality reduction techniques. **RAYS** helps plot all the features and points in the dataset and aims to provide information such as the spatial positioning of points or classes, also helping in understanding the closeness of multiple points, clustering of points, ordering of subspaces, etc. This visualization is a novel technique built after exploring major viable options to plot the higher-dimensional data, which included parallel coordinates, circle segments, Hiedi++, etc along with some dimensionality reduction techniques, like Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE) etc., used in visualization which cannot bear high number of dimensions. RAYS also provides an option for implementing and visualizing different forms of data based on various parameters, such as variance and Gini index. The dimensionality reduction techniques do not provide complete information about the data to the users and tend to lose certain information. Also, the current existing techniques like parallel coordinates, circle segments, etc, have certain limitations like overlapping of points, constraints on the number of dimensions, and the space in which the visualization is being plotted is not utilized properly.

RAYS provide a clearer view, with no overlap of points, also catering to a large number of dimensions (approx 20 dimensions), utilizing the space provided for visualization efficiently. RAYS radially visualizes the data, with the help of concentric rings, each having a certain number of sectors signifying the quadrants of a plane. RAYS helps the user answer questions like *(a) In which sector the maximum points lie*, *(b) how are the points classified*, *(c) finding relative spatial positions of the points*, etc.

In this Thesis, we understand the structure and applications of RAYS and how it is beneficial over the other pre-existing techniques to plot high-dimensional data.

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Symbols

α	path loss exponent
B	available bandwidth
β_{AF}	AF gain factor

List of Related Publications

Chapter 1

Introduction

1.1 Introduction

1.2 Problem Description

1.3 Contributions

1.4 Organization of Thesis

Chapter 2

Related Work

High-Dimensional Data Visualization has been a significant area of research, with numerous techniques developed to represent high-dimensional datasets and aiming to preserve the intrinsic structure and relationships with data, facilitating interpretation and analysis. But these techniques have their own limitations, which provide us with a base to build our own visualization, solving some of these limitations. Many researchers have tried different methods to plot higher-dimensional data, be it by using dimensionality reduction techniques or by new visualization methods. Here are some of the pre-existing techniques and visualizations used to plot high-dimensional data. ✓

2.1 Dimensionality Reduction Techniques

There exist many different types of linear and non-linear techniques of dimensionality reduction. Many try to plot higher-dimensional data after implementing these dimensionality reduction techniques to reduce the number of features while ensuring maximum information retention in the data. These are useful when the information loss is not much or bearable. We study these to understand why it is important not to reduce the dimensions of the data and, gain insights on what type or how much information one loses after implementing dimensionality reduction.

Can we qualitatively & quantify this loss.

References in code

There are various kinds of dimensionality reduction techniques like Principal Component Analysis(PCA) (Linear technique), Multidimensional Scaling(MDS), Isomap, Maximum Variance Unfolding(MVU), Multilayer Autoencoder, etc, each of which is curated for a certain type of datasets like the swiss roll dataset, biomedical data etc. These techniques aim to preserve the global or local properties of the data.

- PCA is one of the most used linear dimensionality reduction techniques. It embeds the data into lower lower-dimensional representation of data. PCA is widely used in domains of face recognition, pattern recognition, feature extraction, etc.
- Dimensionality Reduction includes non-linear techniques like Multidimensional Scaling (MDS) maps high-dimensional data into lower lower-dimensional representation while retaining the pairwise distance between data points, helping in maintaining the relative relationships in and the lower-dimensional space. MDS is most used in molecular modeling, fMRI analysis, etc. Reference
- IsoMap, another non-linear dimensionality technique, is a better technique than MDS, focusing on preserving the pairwise geodesic distances between data points. Geodesic distance refers to the shortest distance between two points measured on a curved surface(over a manifold). It employs a graph where the nodes represent a data point and the nodes are connected with k nearest neighbors, and the shortest path between two points is easily found using Djiktra’s or Floyd’s shortest path algorithm. There are certain limitations of the Isomap algorithm that severely impact the performance. list them ✓ [Reference]
- Maximum Variance Unfolding (MVU), another non-linear dimensionality reduction technique, similar to Isomap, defines a neighborhood graph and retains pairwise distances in the resulting graph. MVU is widely applied for sensor localization and DNA micro array data analysis.

2.2 High-Dimensional Visualizations

2.2.1 Circle Segments

Reference

Circle Segments is a pixed-based data visualization built for high number of data points and dimensions. The visualization maps the dimensions as segments of a circle. The screen is limited to 100-5000 data points when visualized but this technique tackles this problem by assigning pixel to each data value. This helps it achieve visualize over 1000000 data points at once. The visualization tool also allows the user to change the order or the segments (dimensions) which help in comparison of various dimensions with each other. The requirement of the visualization is that there must be minimum 3 dimensions in the dataset.

Show visuals

This visualization is necessary for us to be studied as it gives us an idea of how various dimensions can be compared - by reordering the segments and how the number of data points being shown on the screen can ~~be~~ large. While this visualization’s goal remain remain to increase the number of data points being shown to the user at once, this is helpful since ”360° RAYS” aims to display higher number of dimensions and as the dimensions grow the ring size increase substantially. However the maximum number of dimensions is not specified, but it surely remains a limitation since 360○ can be divided into only a limited number of segments to visualize the data points clearly and effectively. Whereas ”360° RAYS” does not have this problem in terms of dimensions but the spaces, which nevertheless can be visualized per pixel.

Not clear

what you say & why you say needs to be clear

2.2.2 Parallel Coordinates

2.2.3 RadViz/RadViz++

2.2.4 Heidi++

2.3 Limitations of existing work